

MAHARANA PRATAP MAHAVIDYALAYA JUNGLE DHUSHAN, GORAKHPUR



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*A PROJECT SYNOPSIS ON "CARESYNC ELITE: ADVANCED HEALTHCARE
MANAGEMENT & TELE-CONSULTATION SYSTEM"*

SUBMITTED FOR THE DEGREE OF BACHELOR OF COMPUTER APPLICATIONS (BCA)

SUBMITTED BY:

HARSHIT VERMA

AYUSH TIWARI

UNDER THE GUIDANCE OF:

ARPITA SINGH

[Head Of Department (HOD)]

GAURAV DWIVEDI

[Assistant Professor]

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Introduction

Title of the Project:

CareSync Elite: A Cloud-Based Healthcare Ecosystem.

Introduction:

In the contemporary era of digitalization, the healthcare sector is undergoing a massive transformation. "CareSync Elite" is an advanced Healthcare Management System designed to simplify the interaction between medical professionals and patients. Traditionally, hospitals rely on manual paperwork and physical queues, which lead to inefficiency and patient dissatisfaction.

This project aims to provide a unified platform where every aspect of healthcare—from booking an appointment to receiving a digital prescription—is handled electronically.

The system focuses on high performance, data security, and a premium user experience (UX), ensuring that even non-technical users can navigate the portal with ease.

By integrating features like real-time token tracking and secure report storage, CareSync Elite sets a new benchmark for digital clinics

Problem Definition

The existing healthcare system in many local clinics and hospitals is plagued by several critical issues:

- **Manual Queue Management:** Patients often have to wait for hours without knowing their exact turn.
- **Data Fragmentation:** Medical history and prescriptions are usually on paper, which is easily lost or damaged.
- **Communication Gap:** There is no direct way for a doctor to broadcast important news (like a change in shift) to all patients at once.
- **Geographical Barriers:** Patients often travel long distances just for a simple follow-up or to show a lab report.
- **Identity Privacy:** Many patients feel hesitant to share sensitive health issues openly at a reception desk.

CareSync Elite solves these problems by providing a digital token system, centralized database, and an anonymous consultation feature.

Objectives of the Project

The primary objectives of CareSync Elite are:

- **To Eliminate Paperwork:** Transitioning from physical files to a secure MySQL database for all patient records.
- **Real-time Appointment Scheduling:** Providing an automated engine that generates token numbers and manages doctor availability dynamically.
- **Digital Prescription Engine:** Allowing doctors to write and upload prescriptions that patients can access anytime from their dashboard.
- **Enhanced Admin Control:** Empowering hospital administrators to verify doctor credentials before they can practice on the platform.
- **Communication Efficiency:** Implementing a broadcast system and a messaging module for seamless doctor-patient interaction.
- **Scalability:** Designing the system in a way that more modules (like Pharmacy or Pathology) can be added in the future.

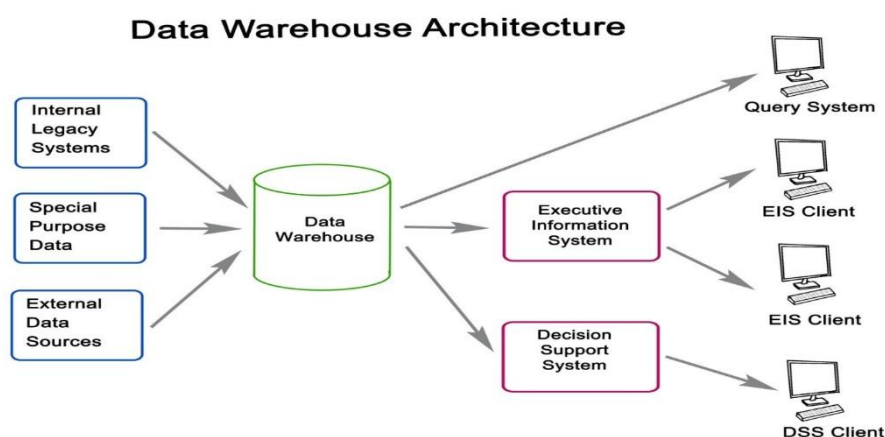
Scope of the Project

- **Internal Scope (What is included):**
 - **Role-Based Access Control (RBAC):** Separate dashboards for Admin, Doctor, and Patient.
 - **Smart Appointment Engine:** Handles date-wise slot booking and token generation.
 - **Medical Report Management:** Secure uploading and viewing of PDF/Image lab reports.
 - **Doctor Verification:** A mandatory approval process by the Admin for every new doctor signup.
 - **Profile Customization:** Doctors can set their clinic address, shift timings, and available days.
- **External Scope (Future Enhancements):**
 - Integration with Payment Gateways (UPI/Netbanking) for consultation fees.
 - AI-based symptom analyzer for preliminary diagnosis.

Proposed System

CareSync Elite is a multi-user, web-based healthcare portal. Unlike traditional systems, it works on a "Zero-Wait" philosophy.

- **For Patients:** They can browse through a list of verified doctors, check their specialization, and book an appointment with a single click. They receive a digital token, which helps them arrive at the clinic only when their turn is near.
- **For Doctors:** They get a streamlined "Today's Queue" where they can see patient details, write digital notes, and close cases efficiently.
- **For Admins:** They act as the backbone, managing the quality of the platform by verifying licenses and managing user accounts.



Methodology / System

Design

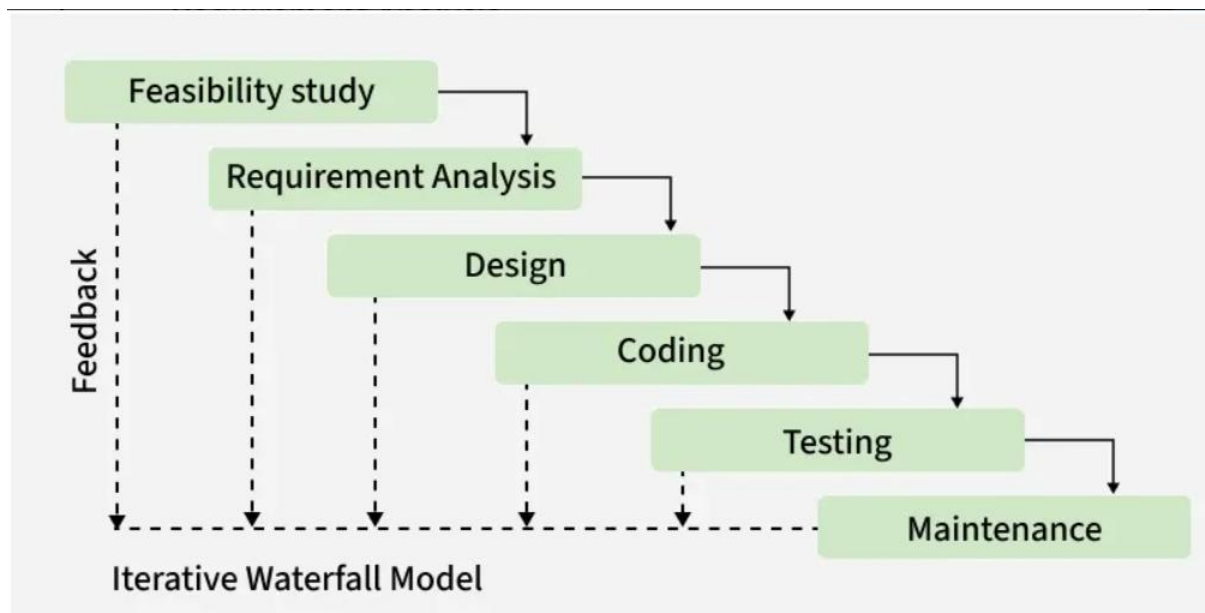
For the development of CareSync Elite, we have adopted the Iterative Waterfall Model. Unlike the traditional linear waterfall, this model includes feedback loops that allow us to revisit and refine previous phases. This ensures that the final healthcare portal is robust, bug-free, and highly secure.

Phases of the Iterative SDLC:

1. **Feasibility Study:** In this initial phase, we evaluated the technical and operational requirements of a digital clinic. We confirmed that using PHP and MySQL on a local XAMPP server was the most feasible way to build a high-performance system for a small-scale team.
2. **Requirement Analysis:** We analyzed the core needs of doctors and patients. Due to the iterative nature, we were able to add advanced features like "Anonymous Mode" (Identity Masking) mid-way through the project to ensure patient privacy for sensitive health issues.

3. **Design:** We created the ER-Diagram and Database Schema. When implementation hurdles arose (such as column name mismatches like problem vs. problem_desc), the model allowed us to go back to the Design phase and update our tables immediately.
4. **Coding:** The project was developed in modules. We first coded the Patient Dashboard and then, based on performance feedback, iteratively improved the logic for the Doctor's Dashboard to ensure seamless real-time data syncing.
5. **Testing:** This was a continuous loop rather than a final step. We conducted unit and integration testing to identify SQL Exceptions and Fatal Errors, instantly jumping back to the coding phase to resolve them.
6. **Maintenance:** After hosting the project on XAMPP, we conducted multiple user trials. This phase ensures the system stays updated, secure, and continues to provide a premium user experience (UX) without any lag.

SDLC Model (Iterative Waterfall Model):



for choosing Iterative Waterfall:

Flexibility: It allows for changes in requirements as the project progresses (e.g., adding the Video Call feature).

Risk Management: Errors like the "Unknown Column" SQL error are caught and fixed in early iterations, reducing the risk of a total system failure at the end.

Efficiency: It is better suited for a two-person team as it allows for constant improvement.

System Flowchart:

1. Entry Point:

- Start: The user opens the CareSync Elite Web Application.
- Login Page: User enters Email and Password.

2. Authentication (Decision Diamond):

- Check Credentials: The system checks the users table in the database.
- If Invalid: User is redirected back to the Login page with an error message.
- If Valid: The system identifies the User Role (Admin, Doctor, or Patient).

3. Role-Based Redirection:

- Path A (Patient):
 - Accesses Patient Dashboard.
 - Searches for a Doctor/Specialization.
 - Fills the Appointment Form (Problem description, Identity masking).
 - Process: Token is generated and saved in the appointments table.
- Path B (Doctor):
 - Accesses Doctor Dashboard.
 - Views "Today's Queue" (Fetched from the appointments table).
 - Selects a Patient -> Writes Medical Advice.
 - Process: Data is saved in the prescriptions table.
- Path C (Admin):
 - Accesses Admin Panel.
 - Verifies new Doctors or manages existing Users.
 - Process: Posts Global Announcements to the announcements table.

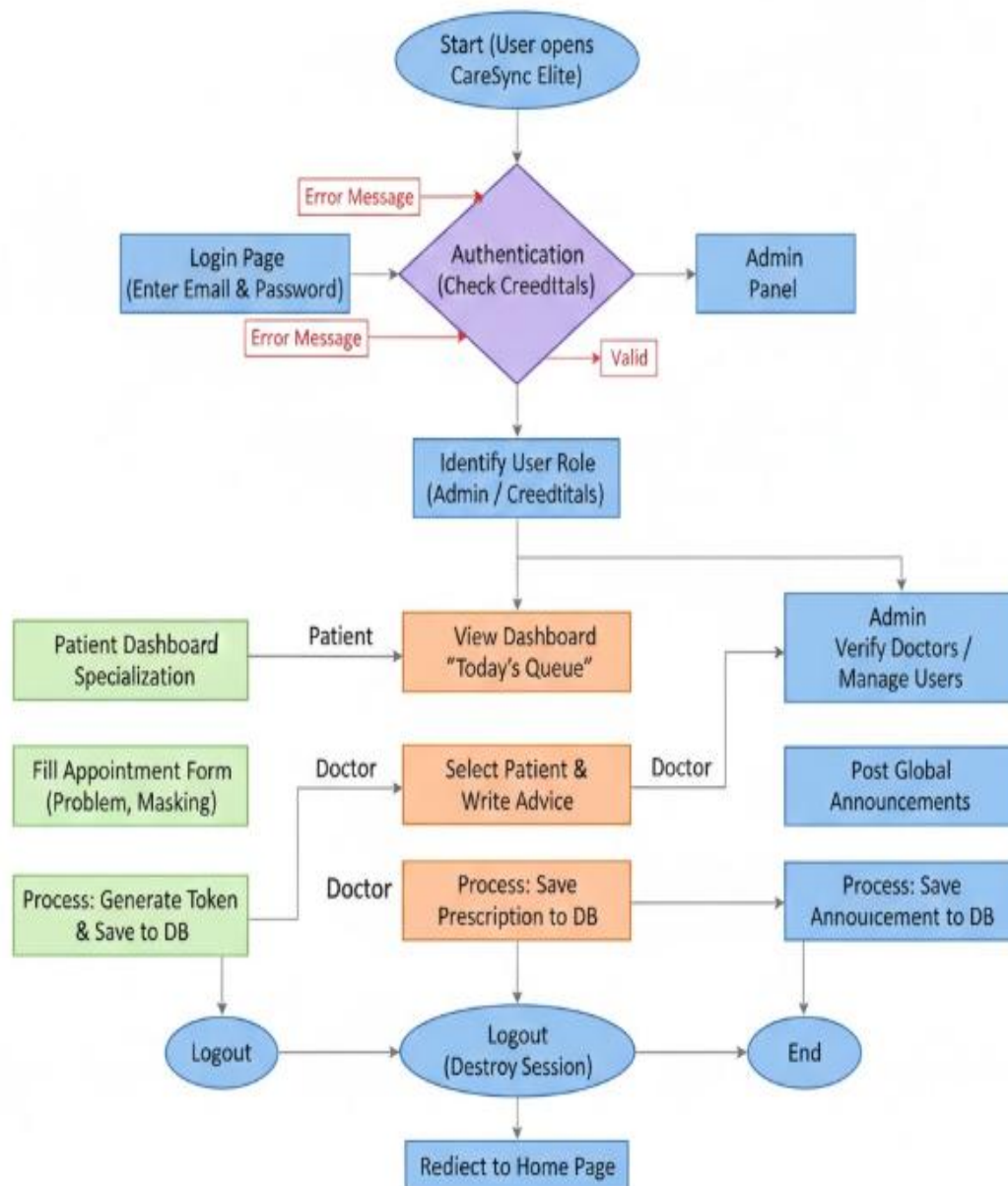
4. Exit Point:

- Logout: The session is destroyed, and the user is redirected to the home page.
- End.

System Flowchart:

CareSync Elite: System Flowchart

Iterative Waterfall Model - Role-Based Access



Database Design

caresync_new consists of several entities & attributes:

Users:

(id,name,gender,age,email,password,role,license_no,specialization,clinic_address,available_days,shift_time,available_time,phone,status).

Appointments:

(id,patient_id,doctor_id,app_date,problem_desc,token_number,status,report_file,hide_identity,token_no,date,issue).

Feedbacks:

(id,patient_id,feedback_text,created_at).

Messages:

(id,appointment_id,sender_id,message,timestamp).

Prescriptions: (id,app_id,medicine_name,advice,report_file).

Reports:

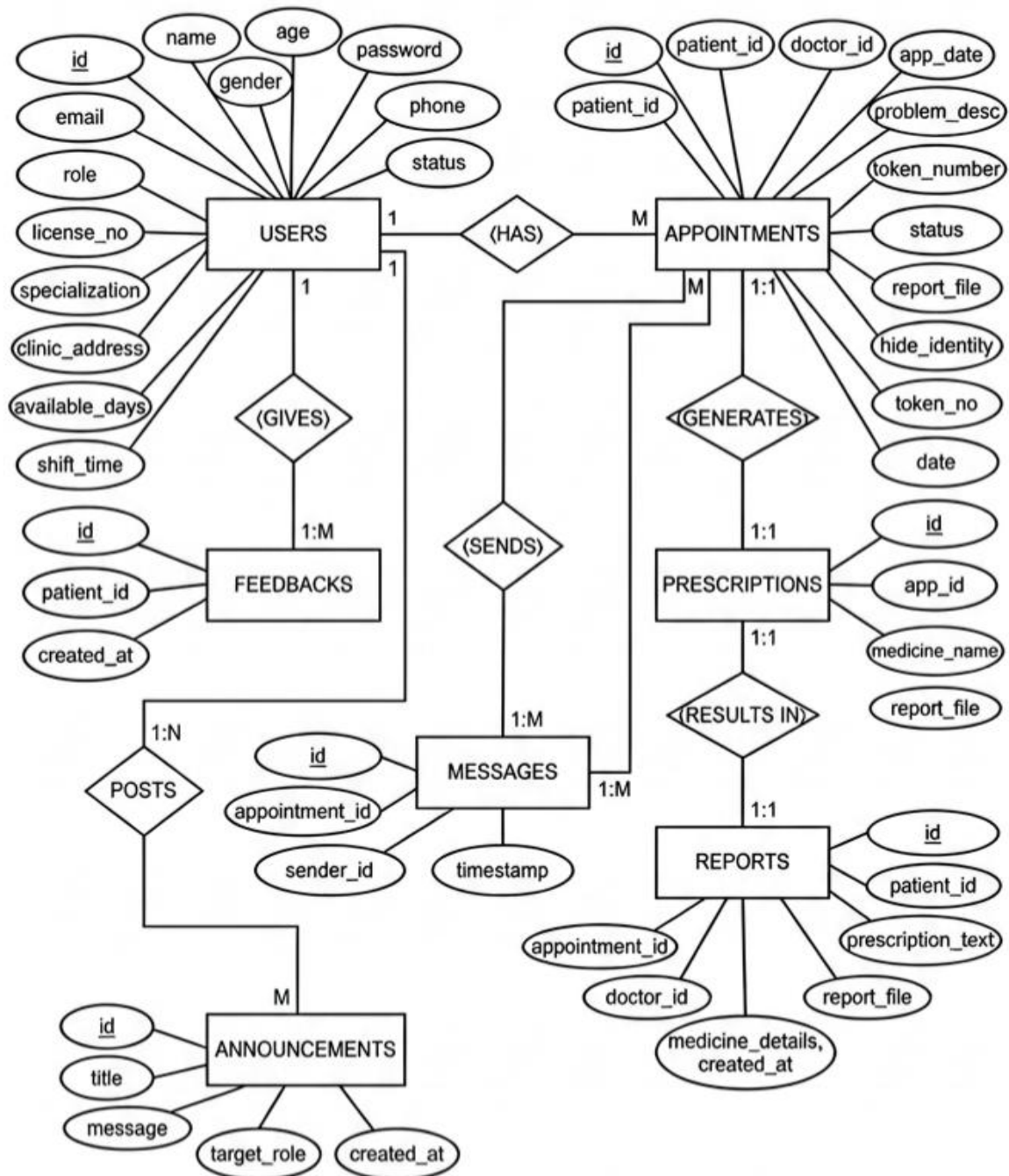
(id,appointment_id,patient_id,doctor_id,prescription_text,medicine_details,report_file,created_at).

Announcements:

(id,title,message,target_role,created_at).

ER-Diagram:

CareSync_New Database Design - Entity Relationship Diagram



Tools & Technologies Used

- **Language:** PHP (Hypertext Preprocessor) - Used for server-side logic and database interaction.
- **Database:** MySQL - Used for storing user credentials, appointment logs, and medical records.
- **Frontend:** HTML5, CSS3, and JavaScript - Used for creating a responsive and animated user interface.
- **Styling Libraries:** FontAwesome (for icons), Animate.css (for smooth transitions), and Google Sans/Poppins fonts.
- **Server Environment:** XAMPP (Apache Server & MariaDB).
- **Development Tool(IDE):** VS Code (Visual Studio Code).

Hardware & Software Requirements

- **Hardware Requirements:**

- **Processor:** Intel Core i3 or higher (for smooth server handling).
- **RAM:** Minimum 4GB (8GB recommended).
- **Hard Disk:** 500MB of free space for database and files.
- **Internet:** For loading external CSS/JS libraries.

- **Software Requirements:**

- **Operating System:** Windows 10/11.
Browser: Google Chrome / Microsoft Edge.
- **Web Server:** Apache (via XAMPP/WAMP).
- **Database Manager:** phpMyAdmin.
- **Languages:** PHP 7.4 or 8.x

Feasibility Study

A feasibility study is conducted to determine whether the proposed system is practical and beneficial for implementation. The following aspects are considered:

- **Technical Feasibility:** The system will be developed using PHP and MySQL, which are widely used and reliable technologies for web development. The application does not require high-end hardware, so it can run smoothly on standard computers used in most clinics.
- **Economic Feasibility:** The development cost of the project is very low because it uses open-source tools such as XAMPP, PHP, and Visual Studio Code. Therefore, the overall implementation cost is minimal and affordable for small clinics.
- **Operational Feasibility:** The system is designed to be simple and user-friendly, allowing clinic staff and patients to use it easily. Only minimal training will be required for staff to operate the system efficiently.

Project Schedule / Timeline

Gantt Chart Table:

Phase / Activity	Month 1 (W1-W4)	Month 2 (W5-W8)	Month 3 (W9- W12)	Month 4 (W13-W16)
Requirement Analysis & SRS				
System Design (ERD & DFD)				
Frontend & Backend Coding				
Testing & Debugging				
Final Documentation & Submission				

- **Month 1 (W1-W4):** Literature survey, Requirement Analysis, and SRS preparation.
- **Month 2 (W5-W8):** Database Schema design (ERD) and UI Wireframing.
- **Month 3 (W9-W12):** Backend development (Login, Dashboards, and Appointment Logic).
- **Month 4 (W13-W16):** Integration of Prescription Engine, System Testing, and Final Submission.

Expected Outcome

The final product will be a fully functional web application that allows patients to book appointments quickly and easily. The system will enable patients to schedule an appointment within 60 seconds, while doctors will be able to manage their daily queue efficiently without manual intervention. All medical records will be stored digitally for easy access and better management.

Key Outcomes:

- Reduced waiting time by approximately 60%.
- 100% paperless medical record management system.
- Improved transparency and communication between doctors and patients.
- Efficient appointment scheduling and queue management.
- Easy access to patient medical history and reports.

Conclusion

CareSync Elite is a comprehensive solution designed to address modern-day healthcare challenges. By using PHP and MySQL, the system provides a scalable, secure, and efficient platform that focuses on improving both patient care and doctor productivity.

This project highlights the importance and effectiveness of web technologies in simplifying healthcare management and improving the overall quality of service.

CareSync Elite is not just an appointment booking system; it is a complete clinic management solution. It helps modernize healthcare services in cities like Gorakhpur by providing digital record management, efficient appointment scheduling, and improved data security.

Overall, the system contributes to creating a more organized, transparent, and patient-friendly healthcare environment.

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Guide Details

Guide Name: ARPITA SINGH

Designation: Head Of Department (HOD)

Guide Name: GAURAV DWIVEDI

Designation: Assistant Professor

Department: BACHELOR OF COMPUTER
APPLICATIONS (BCA)

Institution: Maharana Pratap Mahavidyalaya,
Gorakhpur.

(Signature of Student)

(Signature of Guide)